MAPS FROM AN ACYCLIC COMPLEX INTO A BOUNDED BELOW COMPLEX OF INJECTIVES ARE NULL HOMOTOPIC

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Editorial note. Counted span is the author's own proof environment, source lines 6532-6565, following the statement at 6524-6530.

Editorial note. No omissions: the only reader-check is the substitution of the constructed h^{n+1} into the displayed homotopy equation, routine bookkeeping.

Allowed standard prerequisites (editorial).

- Bounded below target. statement hypothesis; I^bullet bounded below consisting of injective objects. Gives the starting degree n below which the homotopy is already built.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $K^\bullet$ be an acyclic complex. Let $I^\bullet$ be bounded below and consisting of injective objects. Any morphism $K^\bullet \to I^\bullet$ is homotopic to zero.*

Proof. Let $\alpha : K^\bullet \to I^\bullet$ be a morphism of complexes. Note that $\alpha^j = 0$ for $j \ll 0$ as $I^\bullet$ is bounded below. In particular, we can find an n such that there exist $h^j : K^j \to I^{j-1}$ for $j \leq n$ such that $\alpha^j = d^{j-1} \circ h^j + h^{j+1} \circ d^j$ for $j < n$. We will show that there exists a morphism $h^{n+1} : K^{n+1} \to I^n$ such that $\alpha^n = d^{n-1} \circ h^n + h^{n+1} \circ d^n$. Note that

$$\begin{aligned}(\alpha^n - d^{n-1} \circ h^{n-1}) \circ d^{n-1} &= \alpha^{n-1} \circ d^{n-1} - d^{n-1} \circ h^{n-1} \circ d^{n-1} \\ &= d^{n-1} \circ \alpha^{n-1} - d^{n-1} \circ h^{n-1} \circ d^{n-1} \\ &= d^{n-1} \circ (d^{n-2} \circ h^{n-1} + h^n \circ d^{n-1}) - d^{n-1} \circ h^{n-1} \circ d^{n-1} \\ &= 0\end{aligned}$$

Since $K^\bullet$ is acyclic we have $d^{n-1}(K^{n-1}) = \mathrm{Ker}(K^n \to K^{n+1})$. Hence we can think of $\alpha^n - d^{n-1} \circ h^{n-1}$ as a map into I^n defined on the subobject $\mathrm{Im}(K^n \to K^{n+1})$ of K^{n+1}. By injectivity of the object I^n we can extend this to a map $h^{n+1} : K^{n+1} \to I^n$. With this choice the reader checks that we indeed have $\alpha^n = d^{n-1} \circ h^n + h^{n+1} \circ d^n$.

By induction on n we conclude we can find $h = (h^j)_{j \in \mathbf{Z}}$ which forms a homotopy between α and 0 as desired. □

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